Solutions to Review Questions for Lectures 1 and 2

2025-1-23

1. Randomization might be logistically difficult and it might be better to use manual assignment to create similar treatment and control groups if the sample sizes are small.

2. The number of polio cases varied significantly from year to year without any intervention, and so it would be impossible to determine if any decrease in polio cases was due to the vaccine or to this natural variation.

3. You run out of data if you use too many variables in the cross tabulation, and you may not have information on an important variable. You can overcome the first problem in some cases with more data, but you can not overcome the second problem because you cannot possibly collect data on all possible variables.

4. A regression discontinuity design compares people just above and just below some cutoff for a metric of interest, with the idea that people just above and just below differ on meeting the cutoff but are otherwise similar. Suppose that a trade school only admitted students who score above 60% on an aptitude test. A research discontinuity study might assess whether the trade school increases earnings but comparing students who scored just above 60% on the test and were admitted to students who scored just below 60% and were not admitted.

5. This statement suggested that the "treatment" and "control" groups were similar on key variables such as wealth, occupation, and characteristics of the houses.

6. This statement suggested that the assignment of people to water companies was similar to a random assignment, with the people being blind to their assignment.